

Day 1 – gRNA design 1

In this course you will learn how to design gRNAs and for two types of CRISPR/Cas9 editing experiments: whole-gene deletions and precise mutations (see gRNA design session 2).

Part 1 - whole-gene deletions:

We'll design gRNAs and donors to remove entire coding sequences. gRNA design principles are the same as for creating frameshift mutations via NHEJ, meaning you can use what you learn here for making NHEJ indels. The most important considerations are to ensure a valid PAM, minimize off-targets, choose gRNAs that cut within 10% to 75% of the coding region, and select efficient gRNAs (paying attention to GC content, homopolymers).

Step-by-step instructions for gRNA design using Benchling:

1. Open the link to the *galK* target region file on Benchling (https://benchling.com/s/seq-PkUjiZr5cUxIEuXwL19m?item_id=seq_K7jMli551&m=slm-HXWE68fBFxBfSNKybOGF) and copy the sequence into your own folder (this will enable you to edit it).
2. Use the Benchling gRNA design tool to design possible gRNAs withing the *galK* reading frame. Start by selecting the *galK* coding region (click on the annotation, this will select all bases in the coding region).
3. Click on the Design CRISPR icon on the right side of the screen (second icon from bottom, looks like a target) and then click on Design and analyze guides.
4. Enter the following parameters in the popup

Design CRISPR guides: Guide parameters

Design type

☒ Single guide

Wild-type Cas9, single gRNA (higher efficiency)

☐ Paired guides

Double Cas9 nickase, two gRNAs (lower off-target effects)

☐ Guides for "base editing" (Komor et al., 2016)

C → T (or G → A) substitution, no dsDNA breaks

Guide length

20

Genome

ASM584v2 (Escherichia coli str. K-12 substr. MG1655)

Don't see the genome you're looking for? We may be able to import it, [just ask.](#)

PAM

NGG (SpCas9, 3' side)

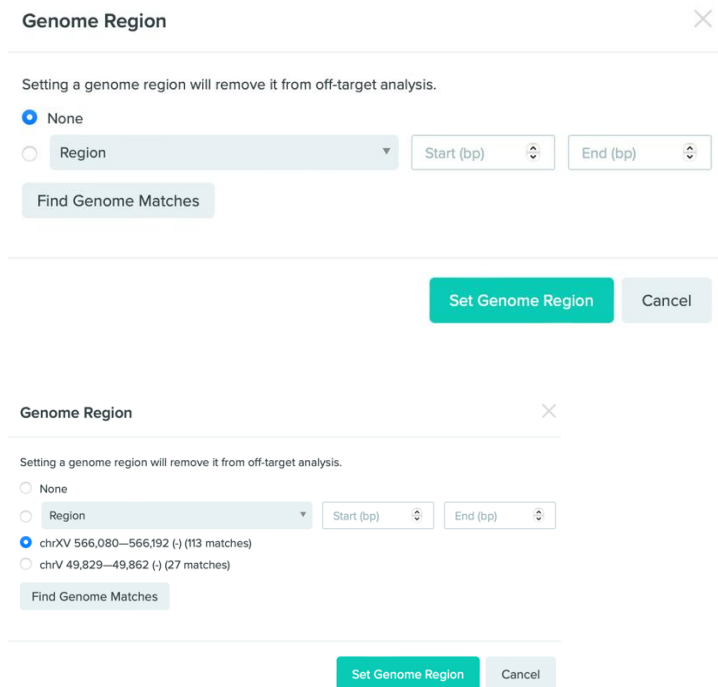
[Show advanced settings](#)

☒ Save these as my default CRISPR settings

Finish

5. **Think about what each of these parameters means and why we chose them.** You can also check what the advanced settings are but leave those at the default settings.
6. Click on Finish. A new tab will appear asking you to select a target region. If you selected the annotation in step 2 the start and end coordinates of the *galK* gene will already be entered and you can just click on '+'. If not, you will need to enter these coordinates manually and then click on '+'.
7. You will have to set a genome region to exclude the target site from the off-target scores provided by Benchling. To do that you can usually click on 'No region set' and then 'Find Genome Matches'. Wait until regions with significant homologies are listed, then select the region with the highest number of matches and press 'Set genome region' (see the

two pasted pictures below). **However, this is not possible for the genome we use, so continue to step 8.**



Genome Region [X]

Setting a genome region will remove it from off-target analysis.

☒ None

☐ Region [Start (bp) [End (bp)]

Find Genome Matches

Set Genome Region Cancel

Genome Region [X]

Setting a genome region will remove it from off-target analysis.

☐ None

☐ Region [Start (bp) [End (bp)]

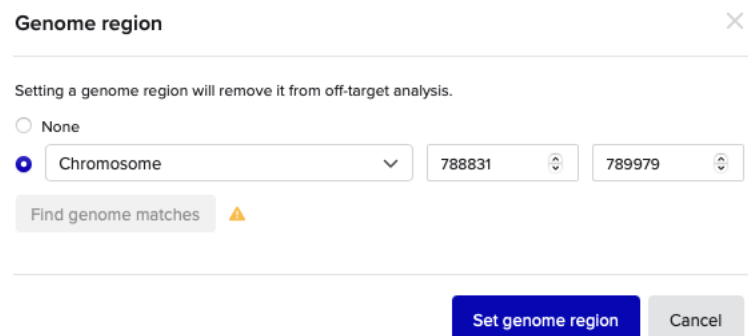
☒ chrXV 566,080–566,192 (-) (113 matches)

☐ chrV 49,829–49,862 (-) (27 matches)

Find Genome Matches

Set Genome Region Cancel

8. If the above is not possible for your genome (it will not be possible for our genome), then enter the coordinates manually and click on Set genome region.



Genome region [X]

Setting a genome region will remove it from off-target analysis.

☐ None

☒ Chromosome [788831 [789979]

Find genome matches ⚠

Set genome region Cancel

9. Select all gRNAs listed in the tab by clicking on the blank square at the top of the list and return to the 'SEQUENCE MAP' tab to see how they align relative to your target mutations. **Look at the different gRNAs and select one that you think would work well, motivate your choice.** Consider the design parameters we discussed. You can see on/off-target scores for each guide by going back to the 'DESIGN CRISPR' tab. To see GC content, select a guide in the SEQUENCE MAP tab and check the grey bar at the bottom of the window.
10. Use CRISPOR (<https://crispor.gi.ucsc.edu>) or CHOPCHOP (<https://chopchop.cbu.uib.no>) to design gRNAs and compare results.
 - For CRISPOR, you can just copy the galk sequence from Benchling (select annotation, cmd +C or ctrl +C). Select a genome in Step2 (E. coli K-12/MG1655), and the appropriate PAM motif in Step3, then submit.

- For CHOPCHOP, you can just enter the gene name and species, along with the application (CRISPR/Cas9 and knock-out)
11. Design gRNA FWD and REV oligos compatible with our cloning vector and paste the sequences below. gRNAs are often ordered as two short, 5' phosphorylated oligos (it is also possible, and cheaper, to phosphorylate them in the lab). The sequences in bold below are the overhangs compatible with gRNA insertion into our BbsI-digested p201 entry vector so **it is important that you add these overhangs to your gRNA sequences**.
- FWD oligo gRNA: 5'-**TAGC**-forward gRNA sequence (20 nt)
REV oligo gRNA: 5'-**AAAC**-reverse complement gRNA sequence (20 nt)
12. **When ordering gRNA oligos, it is important not to include the PAM – why?**